Abhinav Gorantla

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Education

Arizona State University, *School of Computing and Augmented Intelligence* *Jan 2026 – Present*
Doctor of Philosophy in Computer Science

- **Advisor:** Dr. K. Selçuk Candan.
- **Research areas:** Causally informed Machine Learning, Multi-Objective Optimization.

Arizona State University, *School of Computing and Augmented Intelligence* *Aug 2023 – Dec 2025*
Master of Science in Computer Science, GPA: 4.0/4.0

- **Honors:** With Distinction.
- **Thesis:** *Speeding-up of the Non-Dominated Sorting Genetic Algorithm-II by Selective De-Correlation*
- **Advisor:** Dr. K. Selçuk Candan.
- **Coursework:** Artificial Intelligence, Multimedia and Web Databases, Knowledge Representation, Data Intensive Systems for Machine Learning, Statistical Machine Learning, Planning and Learning methods in Artificial Intelligence.

Vellore Institute of Technology, *School of Computer Science and Engineering* *Jul 2019 – May 2023*
Bachelor of Technology in Computer Science and Engineering

- **GPA:** 8.94/10.0
- **Advisor:** Dr. Krishnamoorthy A.
- **Coursework:** Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Networks, Applied Linear Algebra, Artificial Intelligence, Machine Learning, Discrete Math and Graph Theory, Image Processing, Applied Statistics.

Publications

CausalBench-ER: Causally-Informed Explanations and Recommendations for Reproducible Benchmarking
Ahmet Kapkiç, Pratanu Mandal, ***Abhinav Gorantla***, Shu Wan, Ertugrul Çoban, Paras Sheth, Huan Liu, K. Selçuk Candan
In Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25), 2025.
DOI: 10.1145/3746252.3761606

Introducing CausalBench: A Flexible Benchmark Framework for Causal Analysis and Machine Learning *[Best Demo Award]*
Ahmet Kapkiç, Pratanu Mandal, Shu Wan, Paras Sheth, ***Abhinav Gorantla***, Yoonhyuk Choi, Huan Liu, K. Selçuk Candan
In Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM '24), 2024.
DOI: 10.1145/3627673.3679218

Tutorials

CausalBench: Causal Learning Research Streamlined
Ahmet Kapkiç, Pratanu Mandal, ***Abhinav Gorantla***, Shu Wan, Ertugrul Çoban, Paras Sheth, Huan Liu, K. Selçuk Candan
Tutorial at the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2025), 2025.
DOI: 10.1145/3711896.3737598

Research and Professional Experience

Graduate Research Assistant *Tempe, AZ*
EMIT Lab, Arizona State University *August 2024 - (present)*

- **Advisor:** Dr. K. Selçuk Candan.
- **Research areas:** Causally informed Machine Learning, Multi-Objective Optimization, Pareto Optimization.
- Conducting research on optimizing the selection process in multi-objective evolutionary algorithms, specifically Pareto-based methods like NSGA and SPEA.
- Developing an optimized algorithm for efficient Skyline retrieval in relational database systems by leveraging causal structure in the data.
- Collaborating with researchers at CASCADE Lab to maintain and improve `causalbench.org`, a platform dedicated to benchmark causal learning algorithms.
- Collaborated with researchers at CASCADE Lab in developing a novel causally informed recommendation system for `causalbench.org`.

Graduate Services Assistant *Tempe, AZ*
EMIT Lab, Arizona State University *March 2024 - August 2024*

- **Advisor:** Dr. K. Selçuk Candan.
- **Research areas:** Causally informed Machine Learning.
- Supported CASCADE Lab researchers in developing the causalbench-asu Python package and website, establishing an end-to-end benchmarking solution for the causal machine learning community.
- Served as a full stack developer on the CausalBench project, contributing to a comprehensive framework for benchmarking causal machine learning algorithms.

Software Development Engineer Intern *Bengaluru, India*
Webknot Technologies Pvt. Ltd. *April 2022 - June 2023*

- Revamped API endpoints within the Palette project, achieving a notable 30% reduction in response times.
- Engineered a custom plugin for Sisense BI software, enabling the seamless display of geojson data on a GeoJSON layer atop maps rendered via DeckGL.
- Fine-tuned data flow for the DeckGL plugin within Sisense by elevating the efficiency of JAQL queries, ensuring a smoother and more responsive user experience.

Projects

Research Publications Analysis tool

- Architected and developed a research publication analysis web application for ASU, integrating AWS S3, SageMaker, and SES. This tool leverages SCOPUS APIs to fetch research papers and perform text analysis of abstracts.
- Utilized LLM (GPT-4) for advanced feature extraction from scraped text.
- Optimized back-end system performance by reducing server response time by 80% and production costs by 40% through the strategic integration of AWS SageMaker.
- Tools Used: ReactJS, NodeJS, Python-FastAPI, RabbitMQ, MongoDB, AWS S3, AWS Sagemaker, OpenAI API.

Enhancing Diversity in the LLM Modulo Framework through Multi-Response Generation

- Developed the Diversified LLM Modulo framework to address looping and redundancy in the LLM Modulo framework.
- Improved the performance of the LLM Modulo Framework on Planning tasks. Tested my framework on the Google Deepmind Natural Plan benchmark (paper) and achieved a performance improvement of 300% by increasing the diversity of LLM (Large Language Model) Responses.
- Tools Used: Python, OpenAI API.

Multimodal Image Retrieval System using Advanced Feature Analysis and Search Techniques

- Developed a Python-based image retrieval engine encompassing feature extraction from Caltech101 dataset

images, latent semantics computation, clustering, and classification.
  - Employed Locality Sensitive Hashing to index image features, optimizing nearest neighbor searches and ensuring scalability for expansive image datasets.
  - Tools Used: Python.

Teaching Experience

Graduate Teaching Assistant — Arizona State University
CSE 515 Multimedia and Web Databases — Fall 2024

Graduate Teaching Assistant — Arizona State University
CSE 510 Database Management Systems Implementation — Spring 2025

Community Service

Student Volunteer
ACM Knowledge Discovery and Data Mining (KDD) 2025 Conference

Skills

Programming: Python, C++, C, Java, TypeScript.

Machine Learning: PyTorch, NumPy, Pandas, Scikit-learn.

Databases Technologies: MongoDB, SQL, SQLAlchemy.

Cloud Technologies: AWS S3, AWS EC2, AWS Sagemaker, AWS Lambda, Google Firestore, Google Firebase Functions.

Web Technologies: NodeJS (family), ReactJS, FastAPI.

Other: Git, Tensorflow, RabbitMQ.

Leadership

Vice Chairperson — *Vellore, India*
THEP Journalism Club, Vellore Institute of Technology
  - Established and sustained a robust MERN stack application as the cornerstone of the club's online newsletter platform.
  - Pioneered the creation of a podcast for our club, achieving an impressive audience of nearly 800 listeners during its inaugural month.

Editor-in-Chief — *Vellore, India*
Fifth Pillar NGO, Vellore Institute of Technology
  - Oversaw and directed a 50-member editorial team, responsible for the editing and publication of articles on our blog website.
  - Innovated by introducing new content formats like "Law Talks," resulting in a remarkable 50% expansion in the club's social media and community outreach.

[0] References available upon request.